Microsoft

AI for IT Operations
Concept

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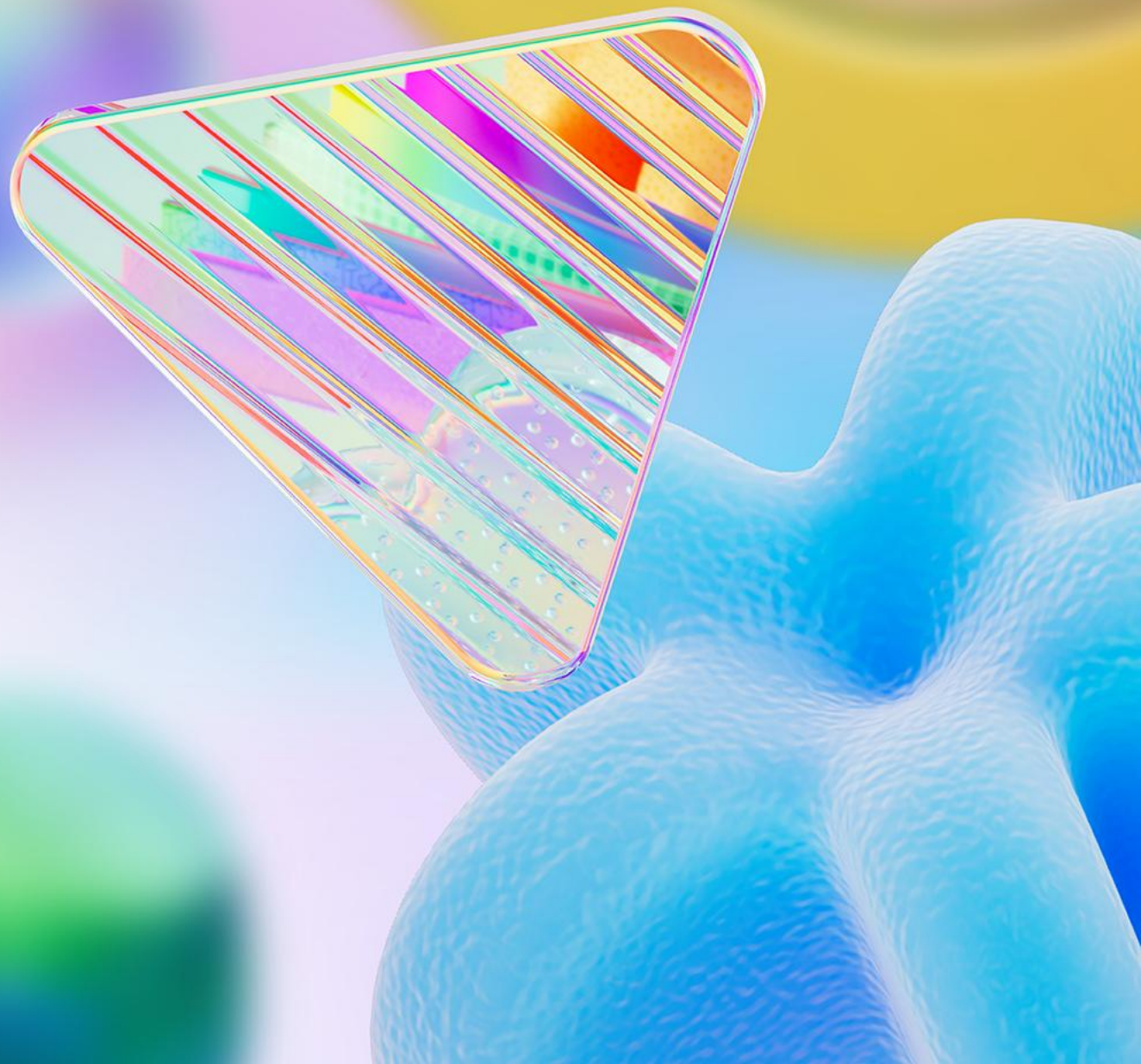
Cloud & AI Solution Engineer

Gianluca De Rossi | LinkedIn

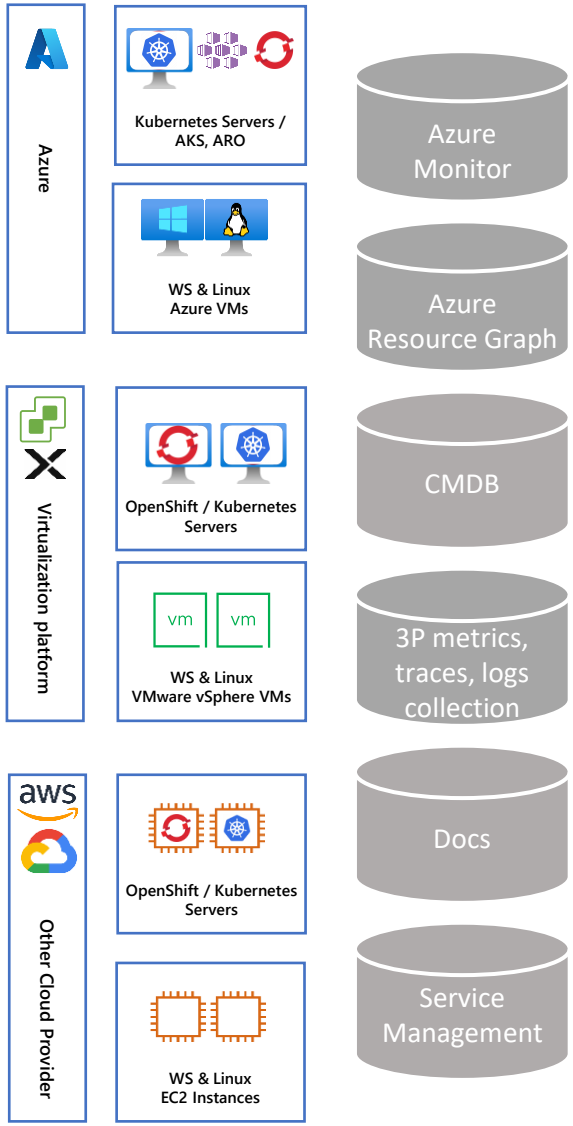


Benefits

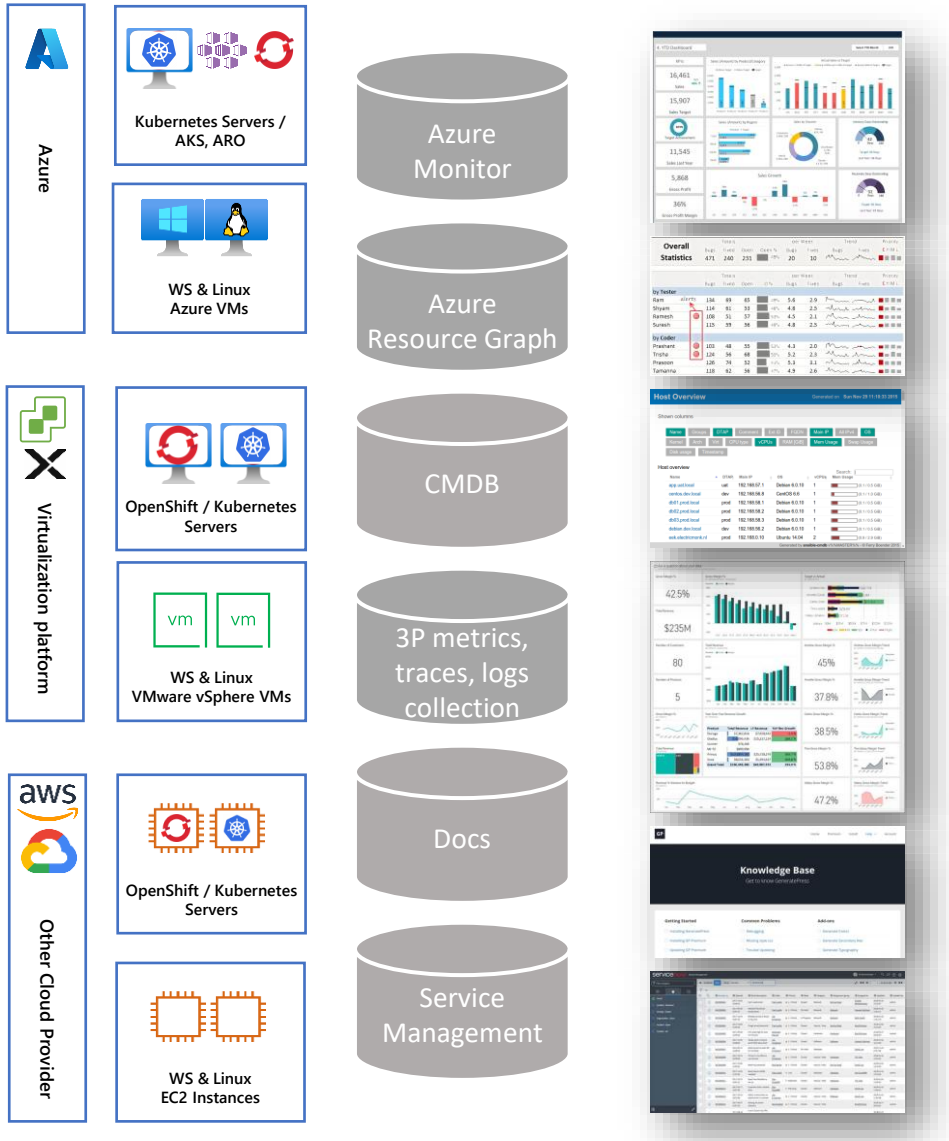
Reduce the load on the Operations

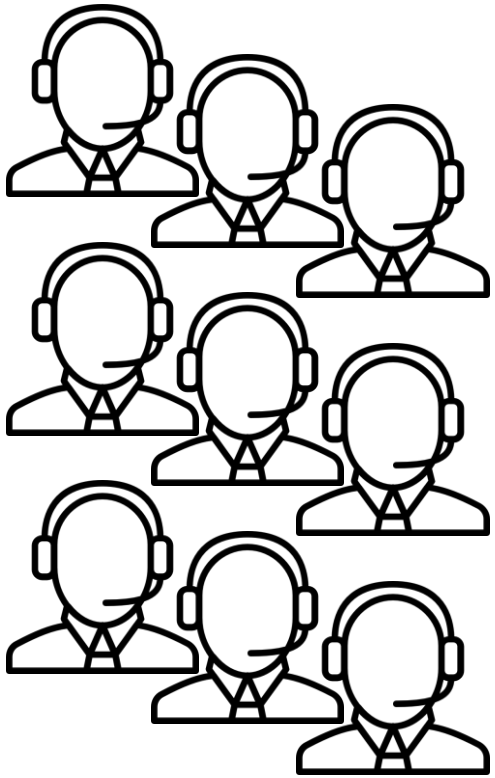
- Automate boring and repetitive tasks
- Analyze large amounts of data faster
- Correlate information between heterogeneous systems more quickly
- Enhance the operators' capabilities

Reference scenario – Typical Ops data sorces

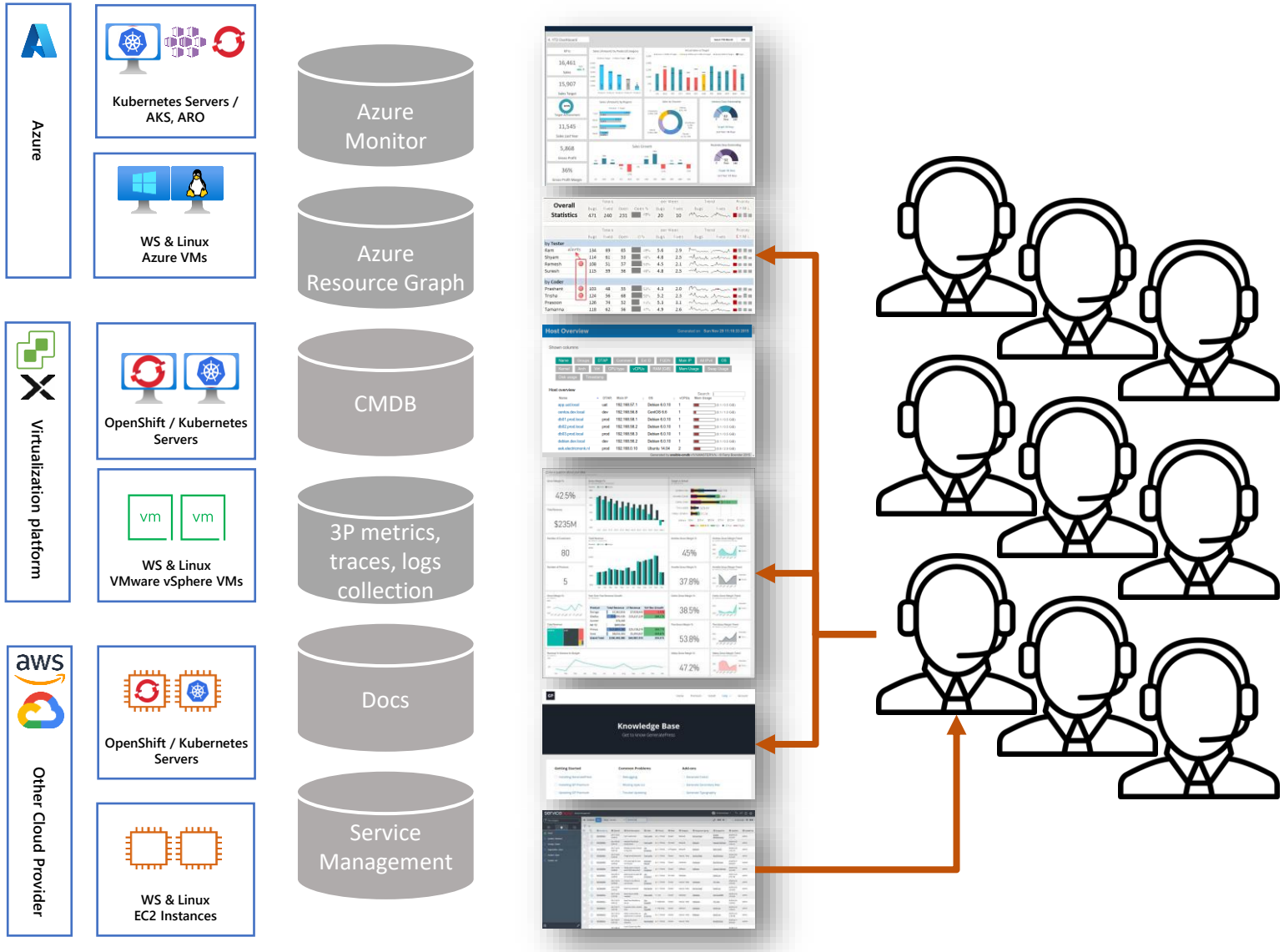


Reference scenario – Ops team organization

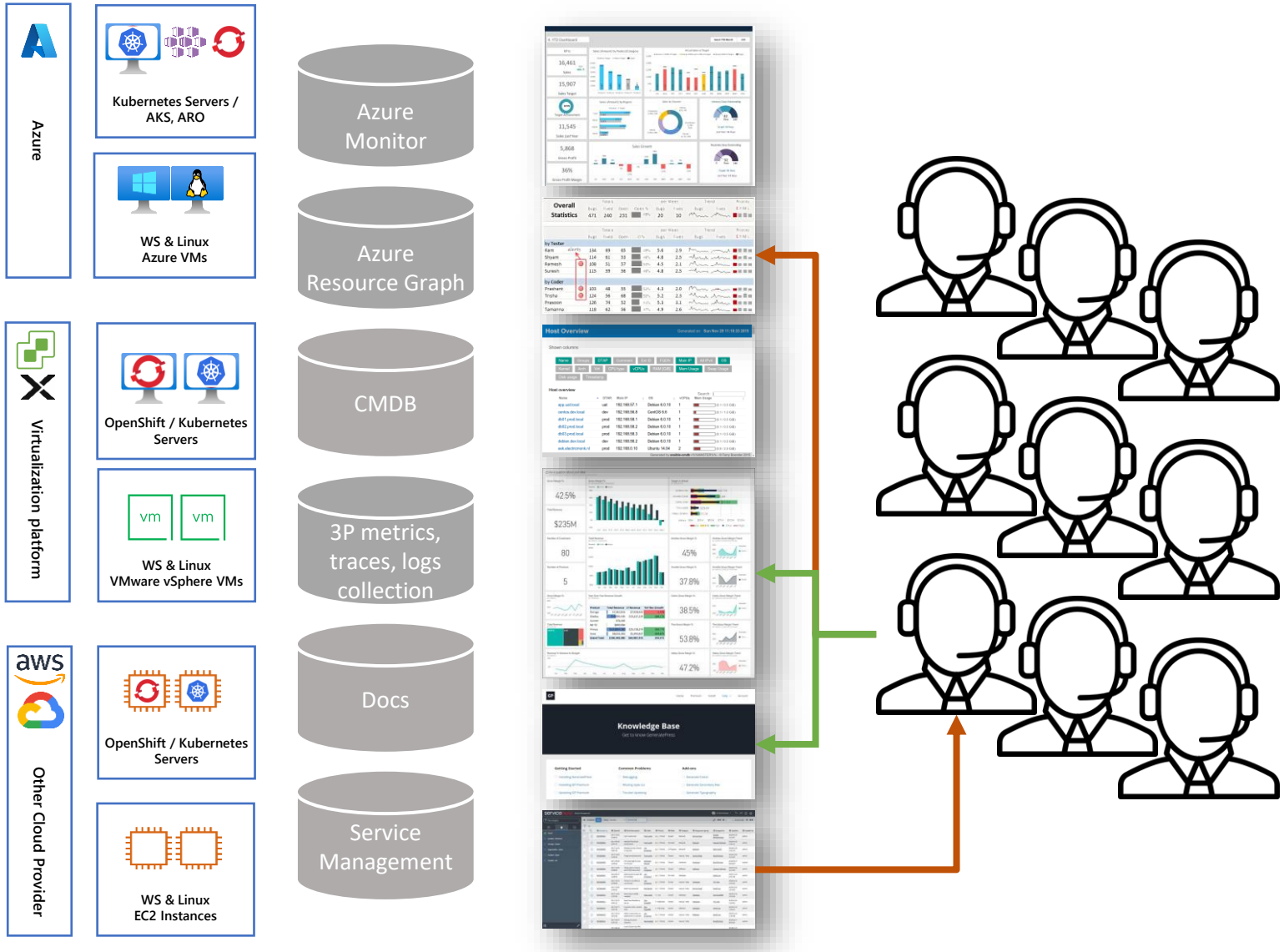




Reference scenario – actual process example



Reference scenario – delegate part of the process to AI



Real-world scenarios

Retrieval

- Applications Deployment Documentation
- Infrastructures Deployment Documentation
- Incidents Documentation
- Metadata Search and Correlation for Servers
- Best Practice Assessments (SQL Server, Windows Server) Query

Process Automation

- Morning Check: receive a report (mail, chat, Teams) on platform status every morning
- Automated Report: compose a report following a checklist
- Enrich ticket information via automated checks
- Smart alerting: add preliminary check to alerts

Reactive / Proactive

- Check provided service quality for a service/application
- Detect anomalies on infrastructure KPIs by platform: Is my core application working fine?
- Find root cause for an anomaly
- Build evidence before a war room or after an incident

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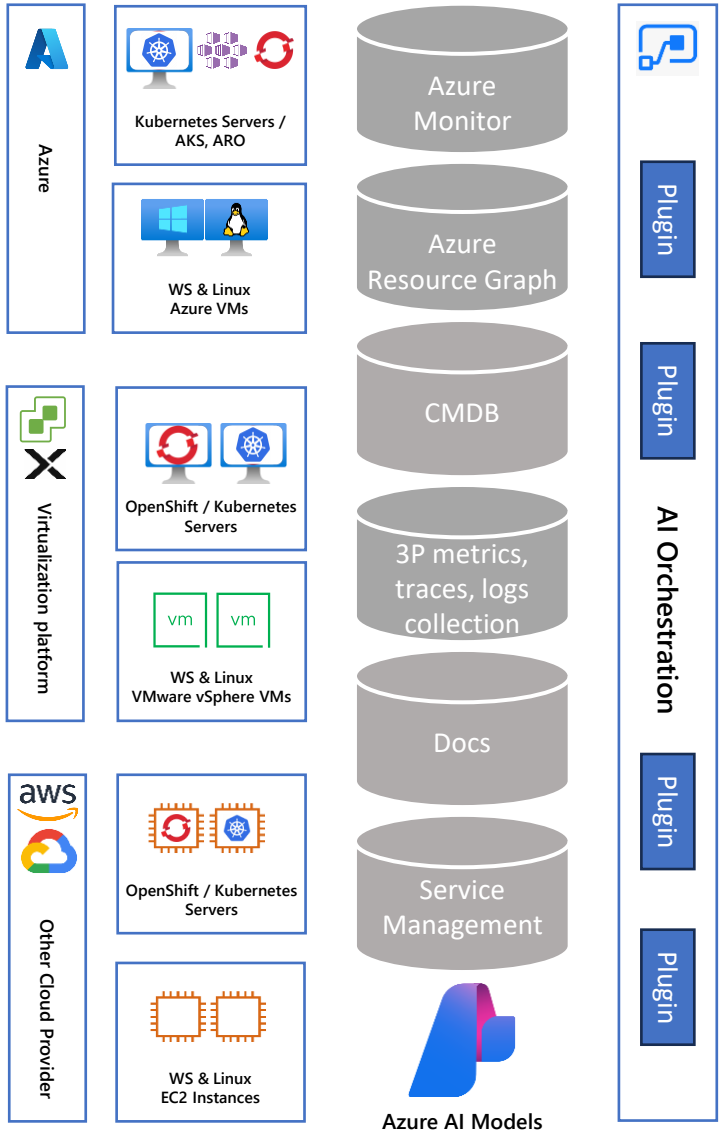
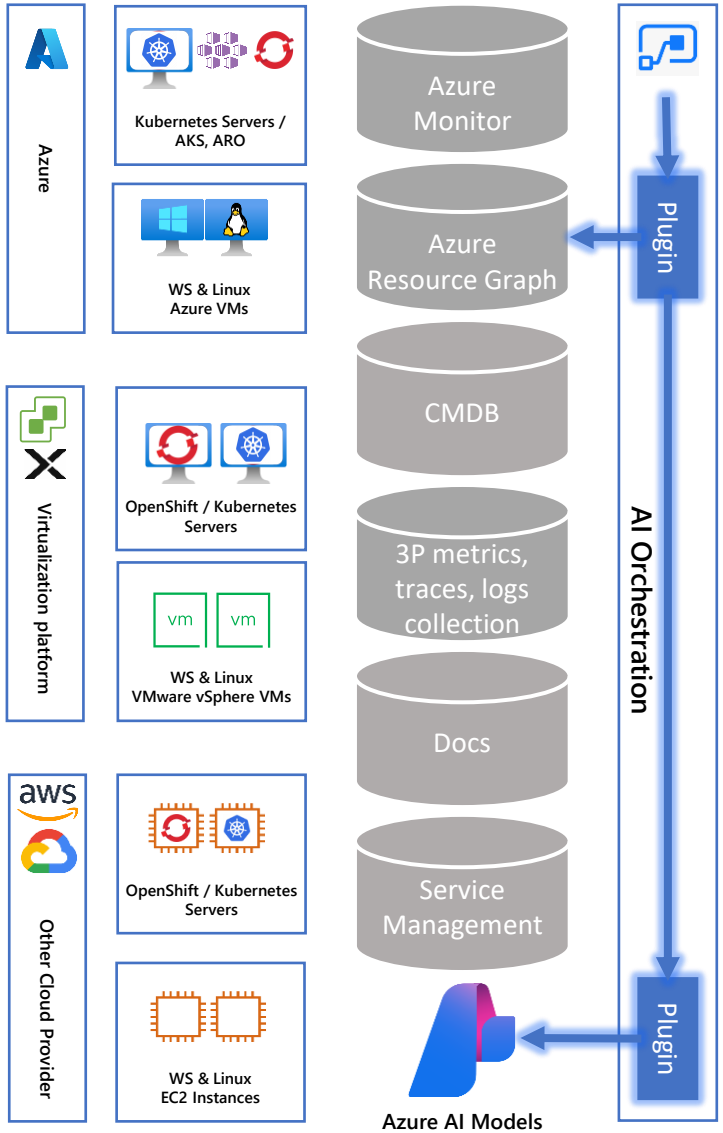
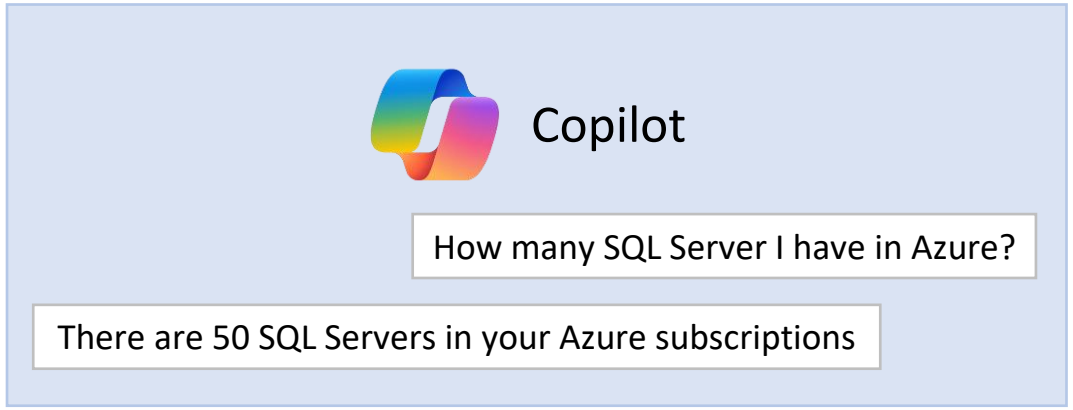
Scenario di Riferimento - AI Orchestration layer

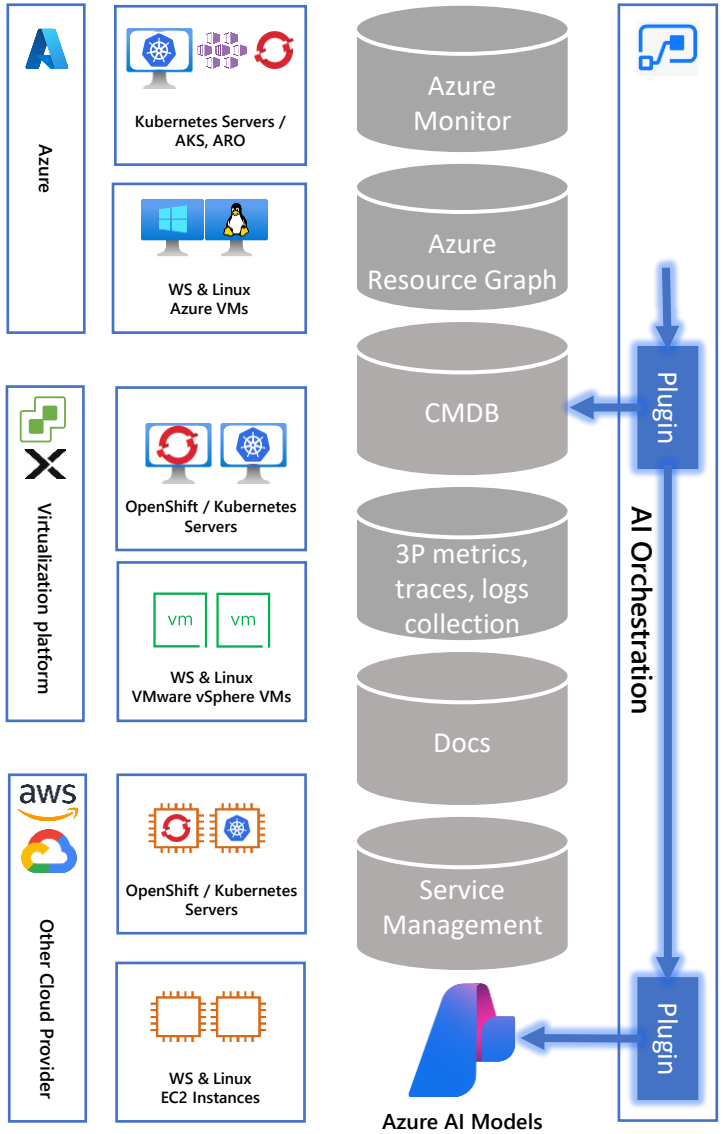


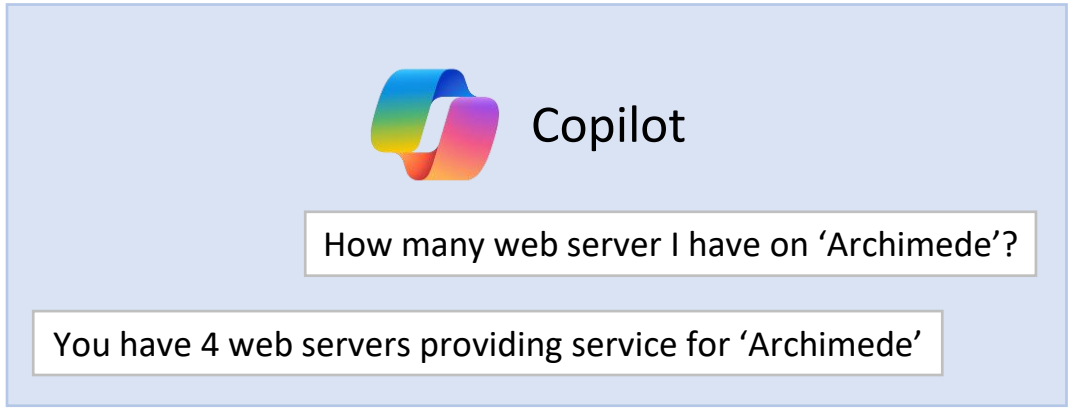
Scenario di Riferimento - Copilot driven flows





Scenario di Riferimento - Copilot driven flows





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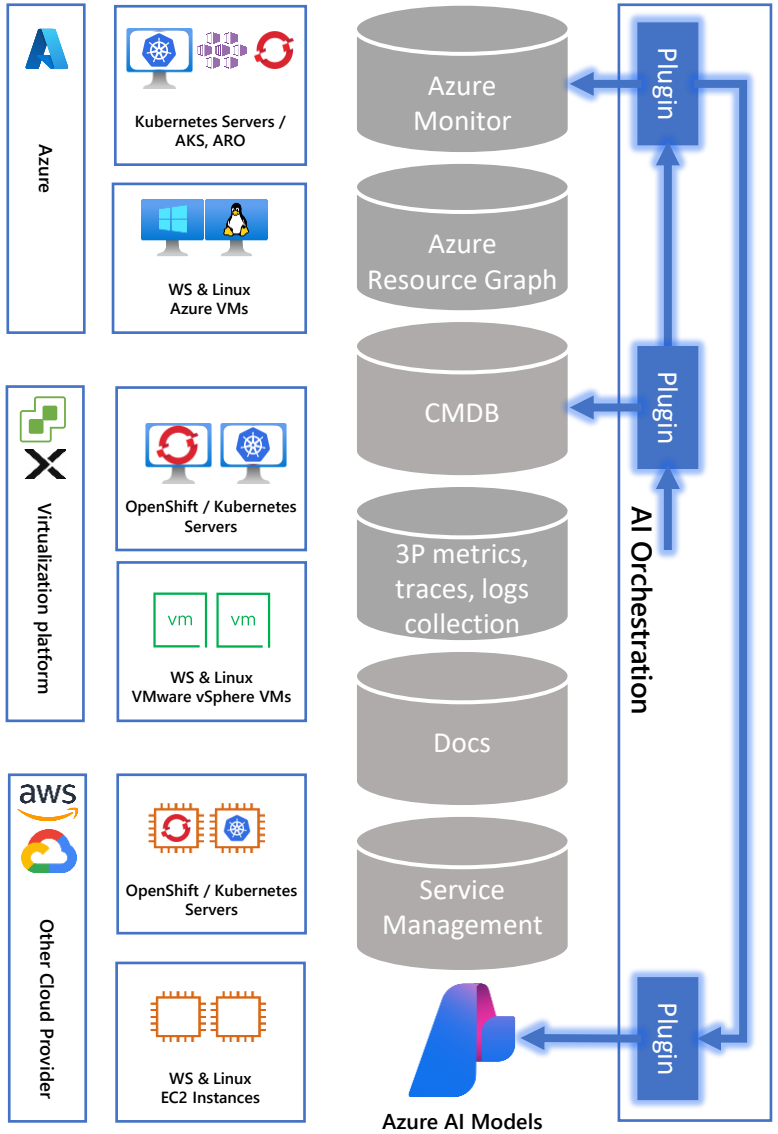

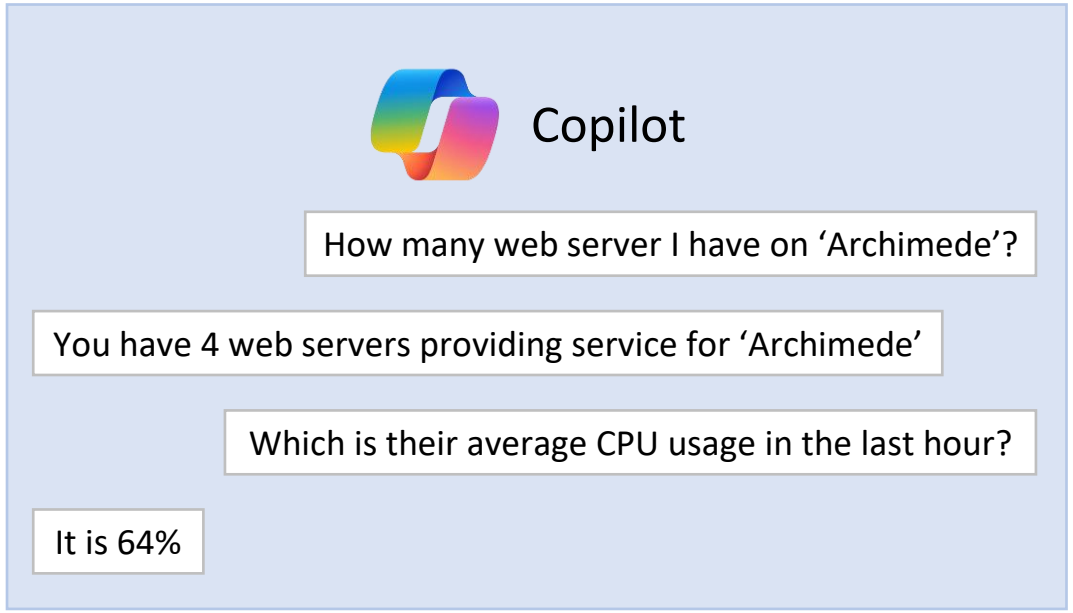

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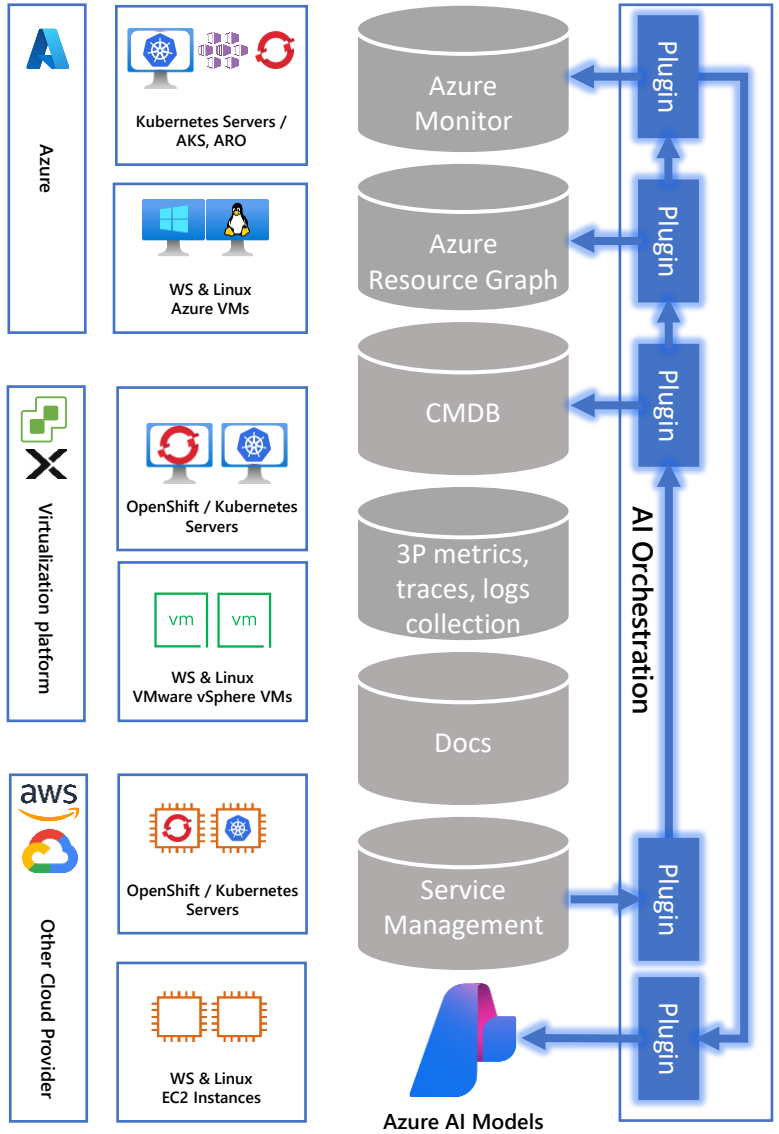
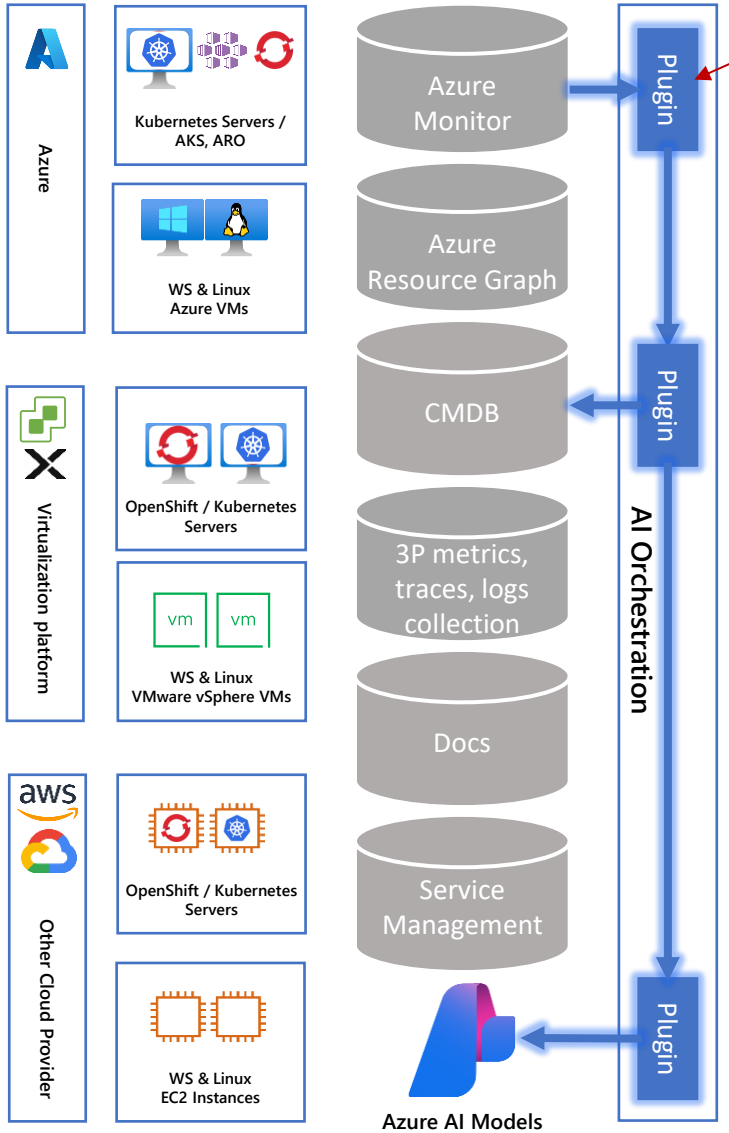
Scenario di Riferimento – Event driven flows



Request ID	1234567
Priority	High
Requestor	James F.
Category	Service not available
Title	Issues on Archimede App
Description	When attempting to log in to the app, users are encountering an error message that states, "Unable to authenticate. Please try again later."
Automated Checks	VMs are all turned on. Patch Management OK. FrontEnd services are running. LDAP connections are down
Operator work notes	

Scenario di Riferimento – Event driven flows



Alert: CPU usage > 70%

Copilot

Alert Description: The alert indicates that the CPU utilization has exceeded the *70%* threshold.

Severity: The severity level of this alert is *Sev1*, which signifies it is of high urgency or importance.

Date and Time: The alert was fired on *January 22, 2025, at 13:07:15 UTC*.

Server Name and Application: The affected server is *WEB-1*, and it is part of the application *Archimede*.

Alerts Count: This is the first alert of this kind, as indicated by an alerts count of 1.

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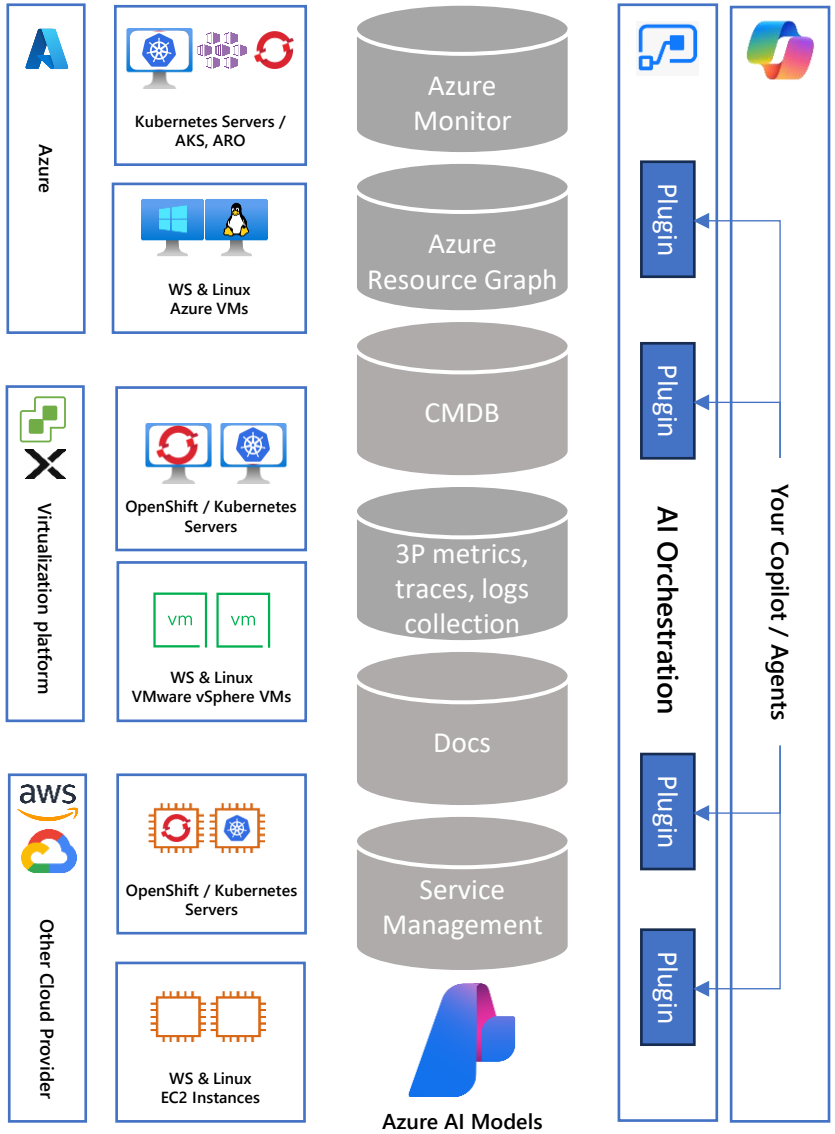
Reference scenario – AI involvement



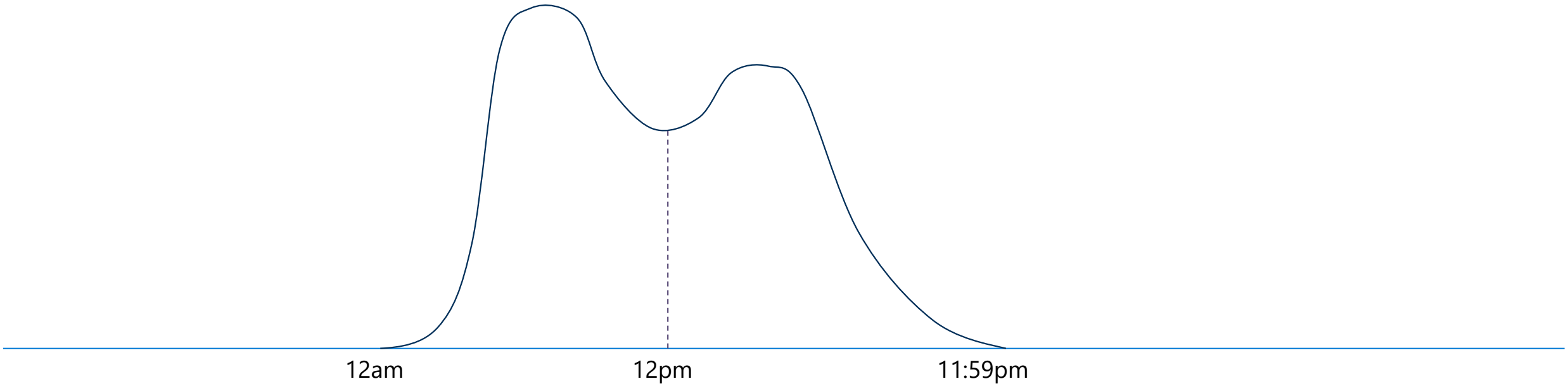
ML capabilities for Ops

- Generate baselines, forecasts and detect anomalies
- Fault localizations
- Capacity management
- Proactive alerting – prevent outages

Generative AI capabilities for Ops

- Find & retrieve documents
- Draft pre-analysis
- Summarize multiple outputs/analysis
- Metadata enrichment, provide contextual information,
- Error explanations / remediations
- Correlations (metrics/traces vs logs, events, etc.)
- Code generation for in-depth analysis

Scenario: Verifica del livello di servizio offerto



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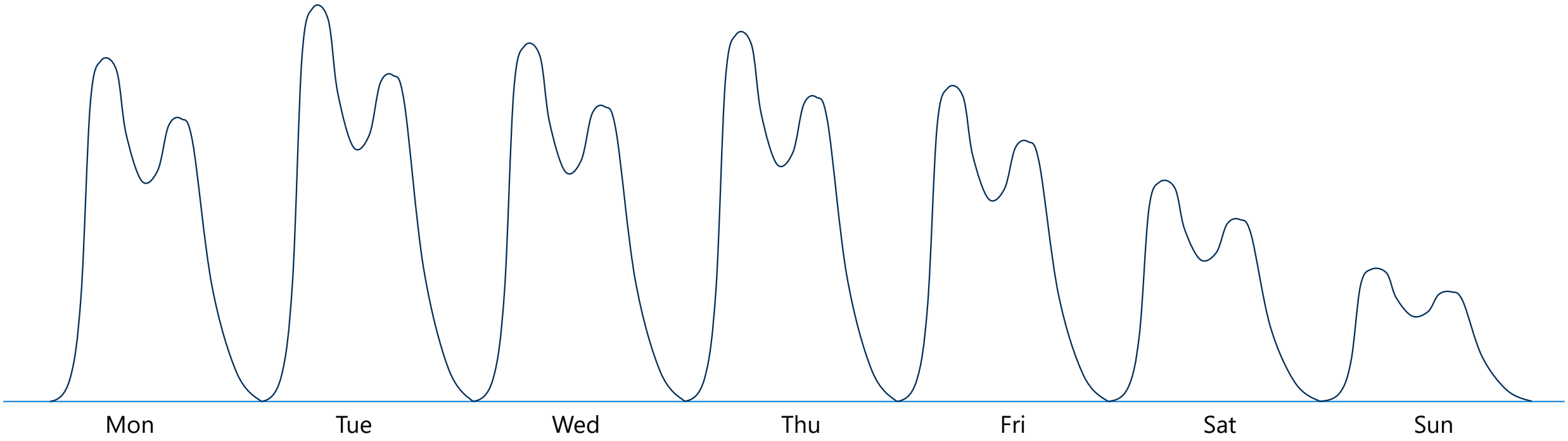

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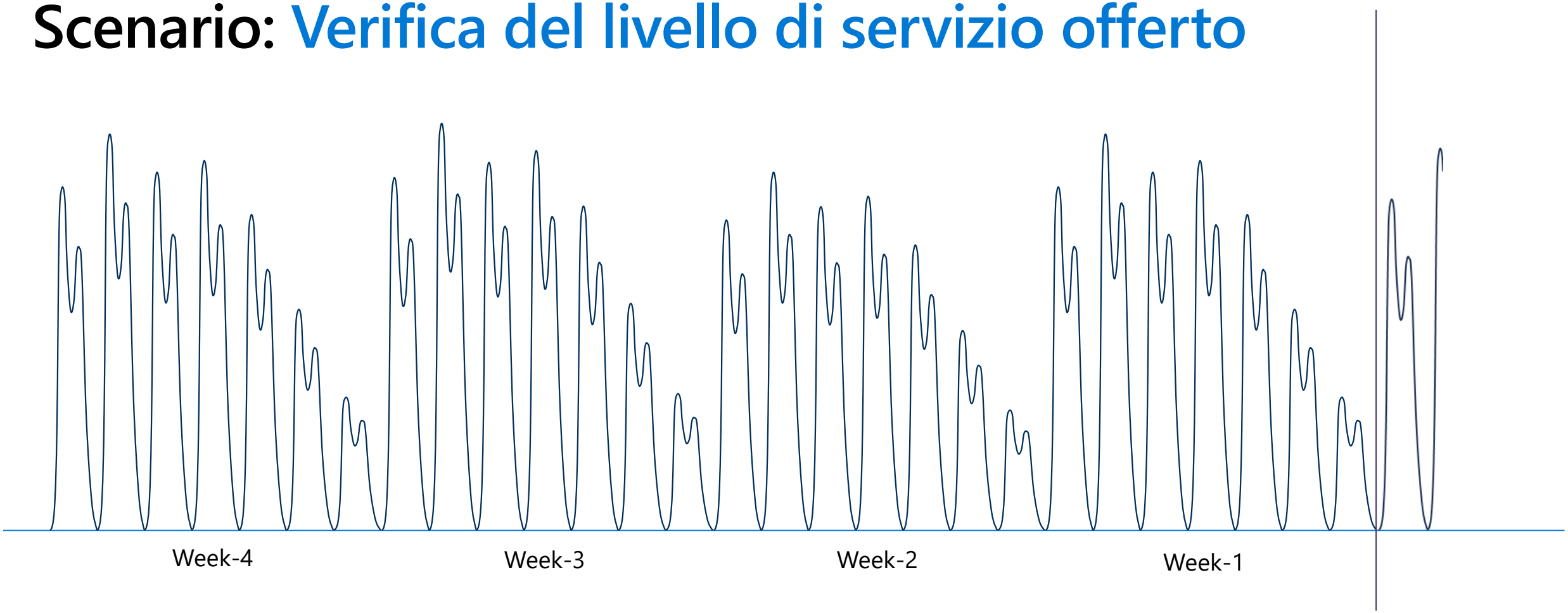

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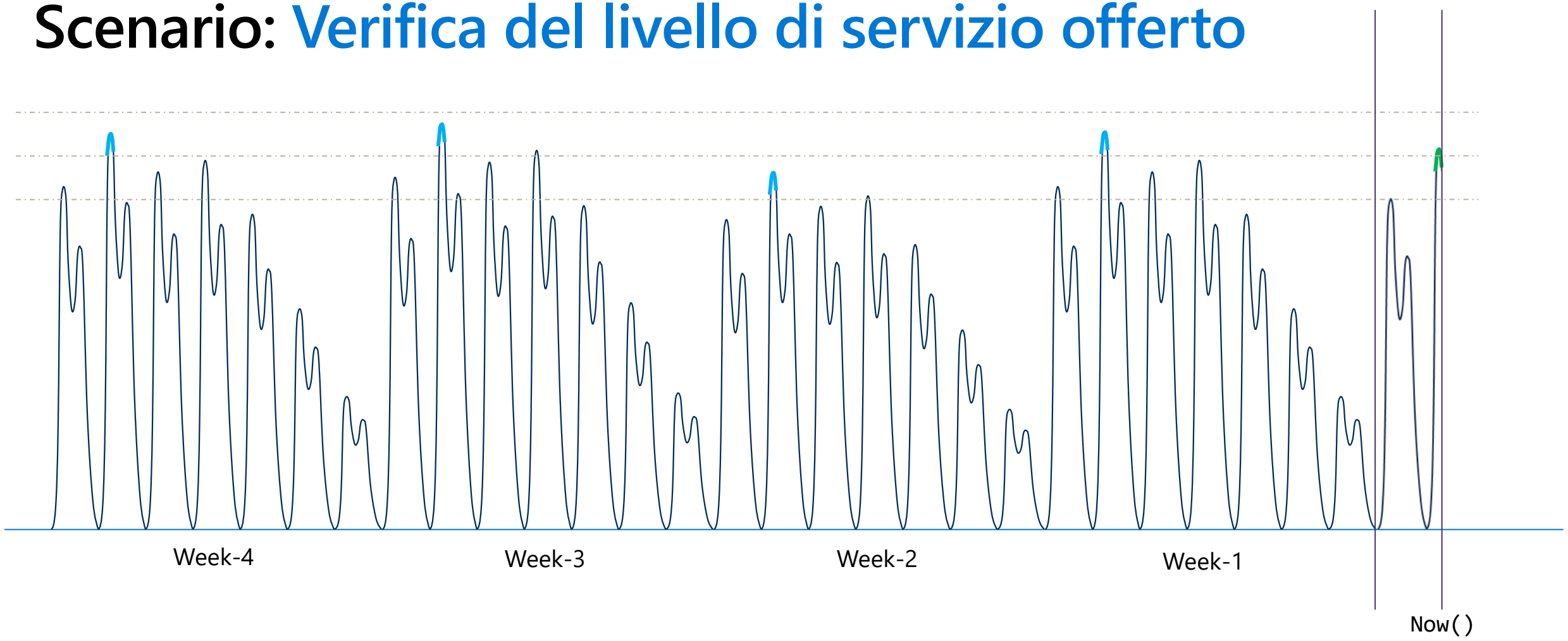

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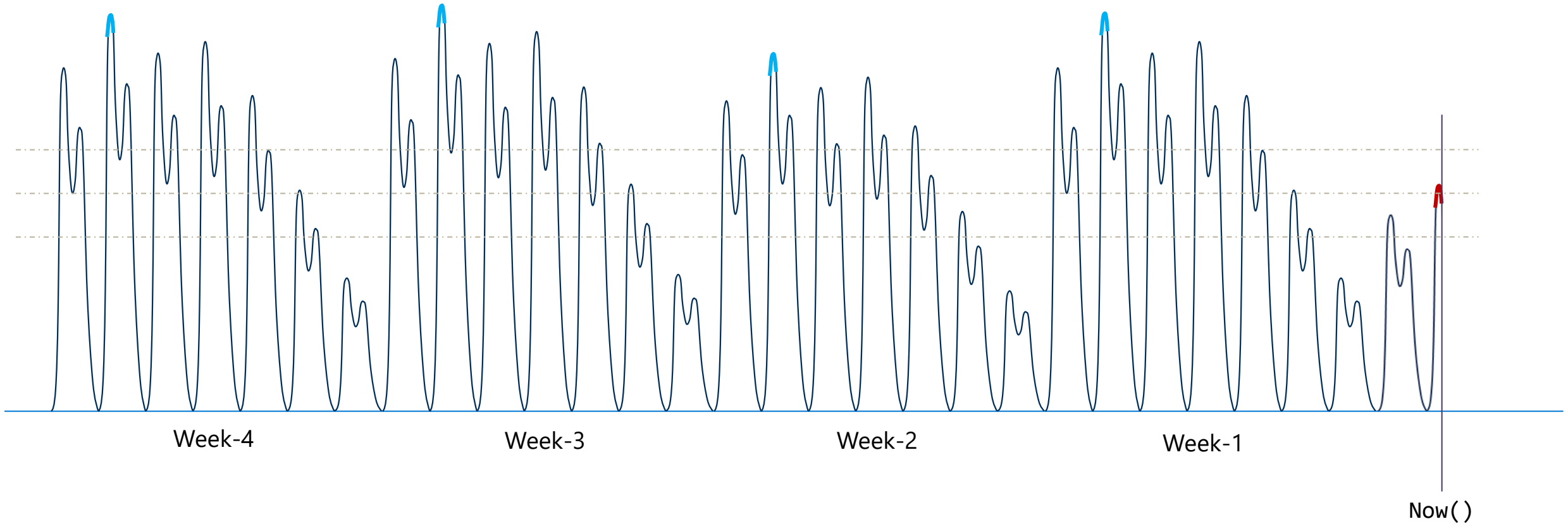
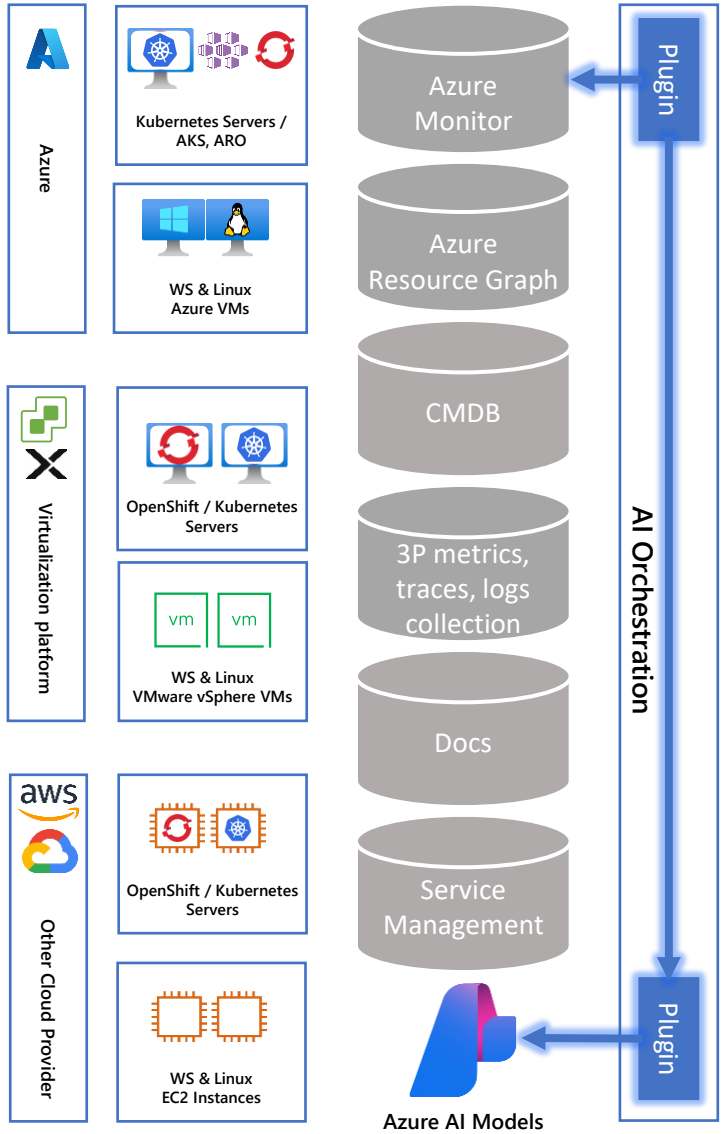


Reference scenario – Flusso Copilot driven no ML



Copilot

Run a health check on Active Directory

Retrieving the behavior for the following KPIs:

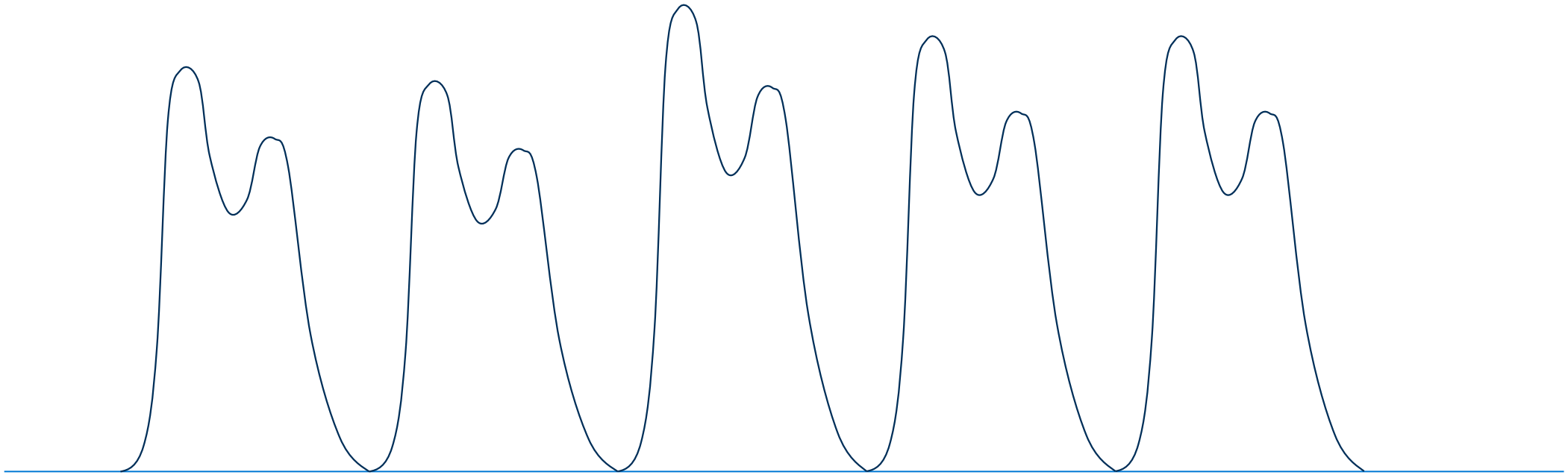
- NTLM + Kerberos authentications
- DNS queries

The following table shows the requested value:

CounterName	Today	Week-1	Week-2	Week-3
NTLM Authentications	75,27	55,64	58,65	63,01
Kerberos Authentications	15,49	14,85	15,02	14,22
Total Query Received/sec	425,88	441,91	431,32	423,05

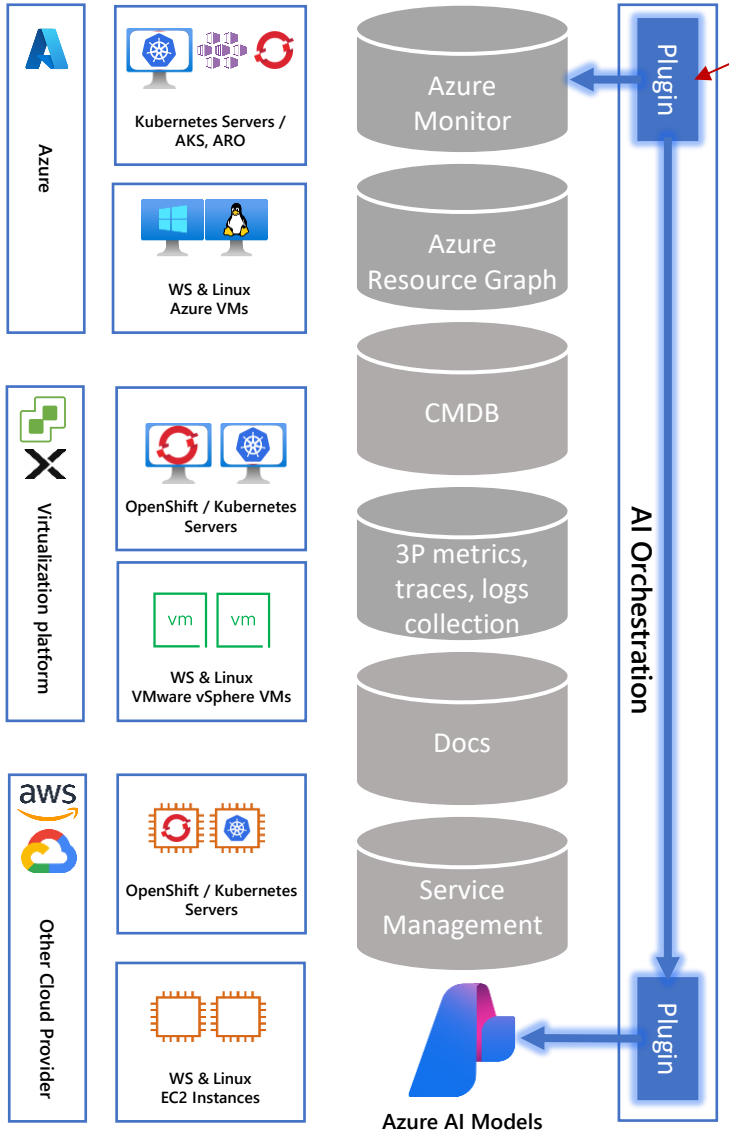
The service level provided today for Active Directory is **Green**.

Scenario: Segnalazione di anomalie del servizio offerto



Scenario: Segnalazione di anomalie del servizio offerto

Scenario di Riferimento – Flusso Copilot driven + ML




```
make_series()
series_decompose_anomalies()
```

Copilot

Do you detect anomalies on my Active Directory?

I am searching for any anomalies that occurred in the last 7 days

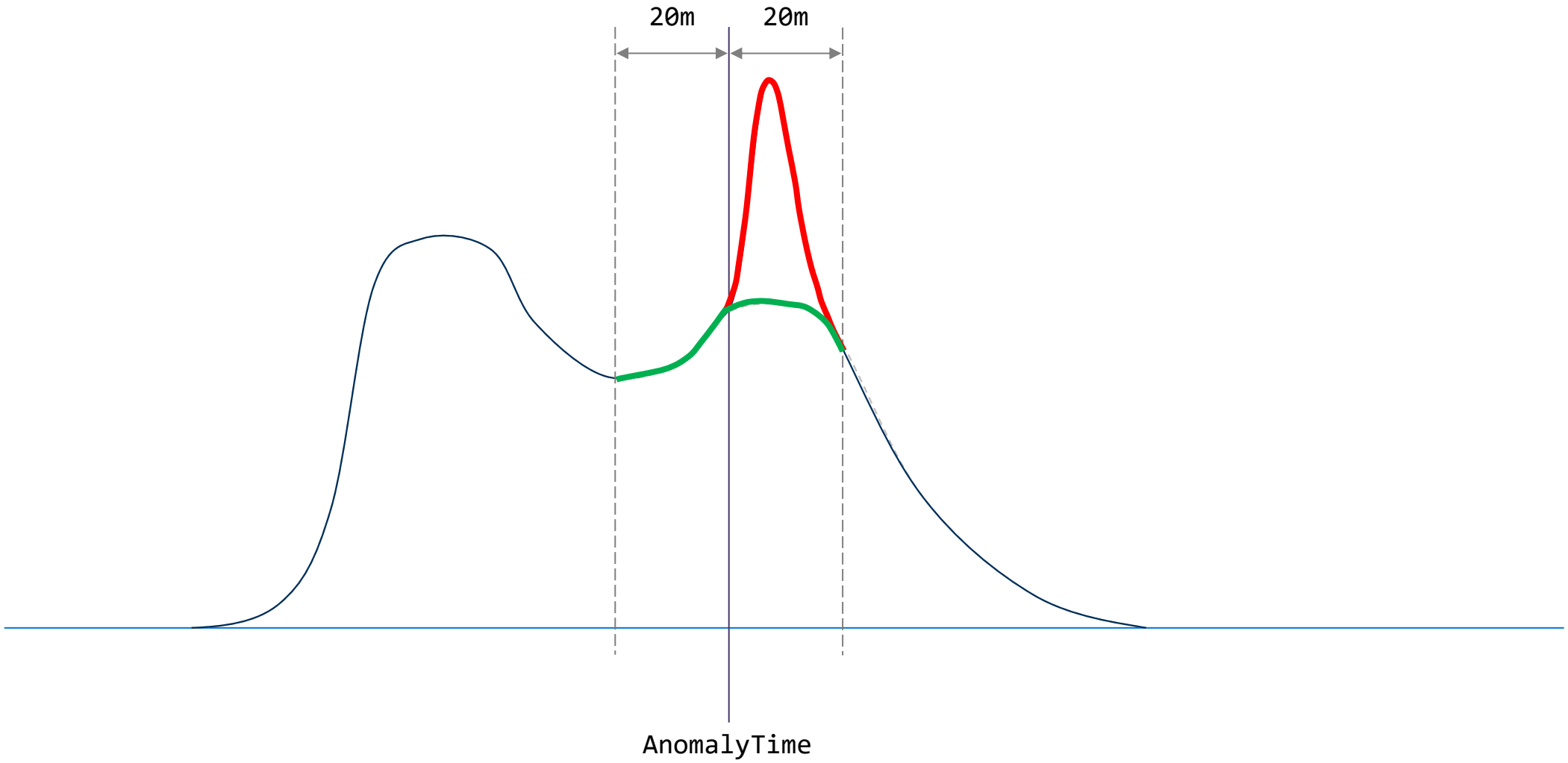
Here is the list of anomalies that occurred over the last 7 days

Timestamp	Domain Controller	EventID	Description
2024-05-04 06:00	DC01	1535	LDAP Error
2024-05-02 20:00	DC04	1216	Error on DC Replica
2024-05-02 20:00	DC05	1216	Error on DC Replica

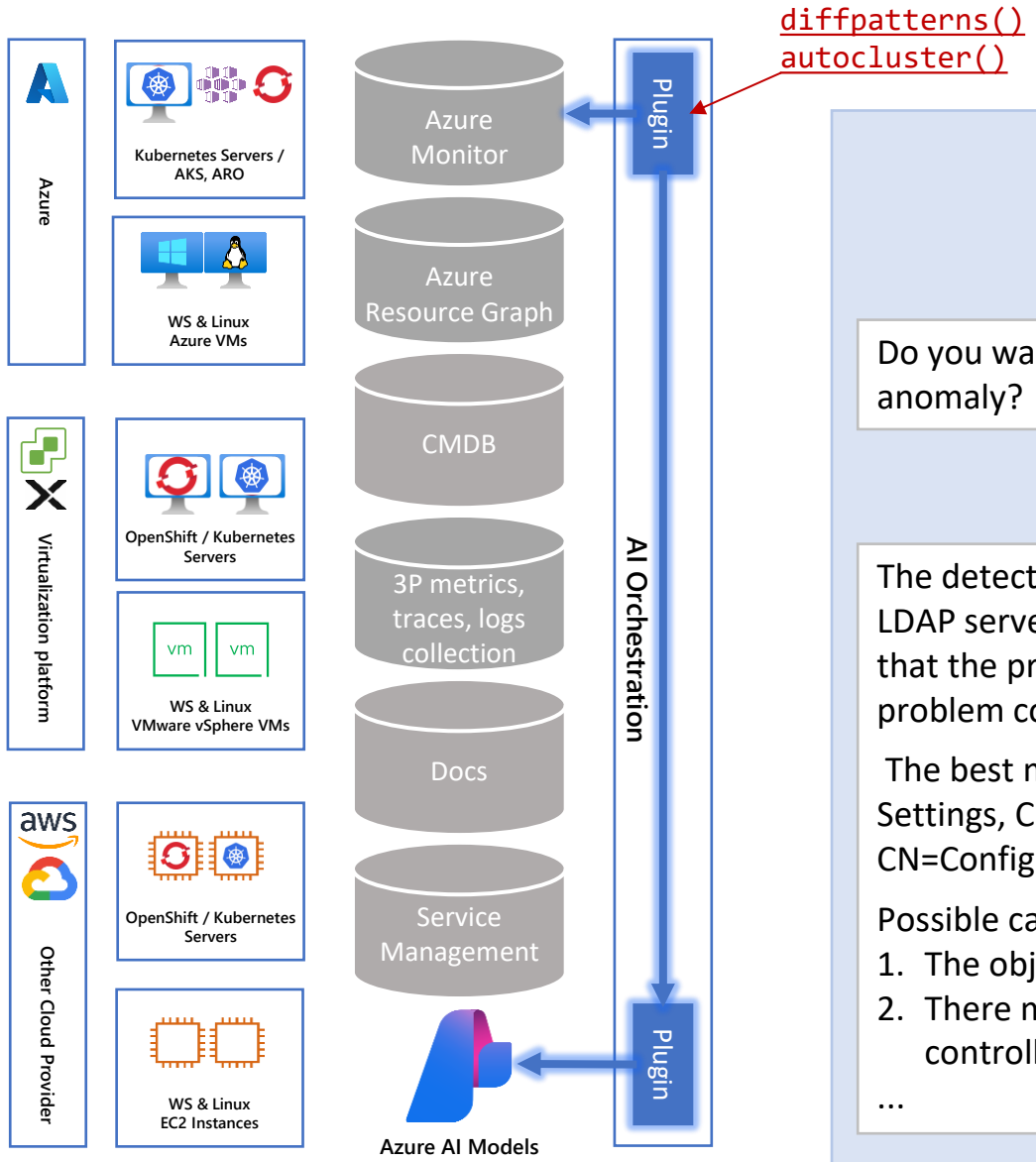
Do you want me to find the root-cause for the most recent anomaly?

Yes

Scenario: Root-cause analysis



Scenario di Riferimento – Flusso Copilot driven + ML



Copilot

Do you want me to find the root-cause for the most recent anomaly?

Yes

The detected EventID is 1535. This event indicates that the LDAP server returned an error. The error description specifies that the problem is caused by an object not found with the problem code 2001 (NO_OBJECT).

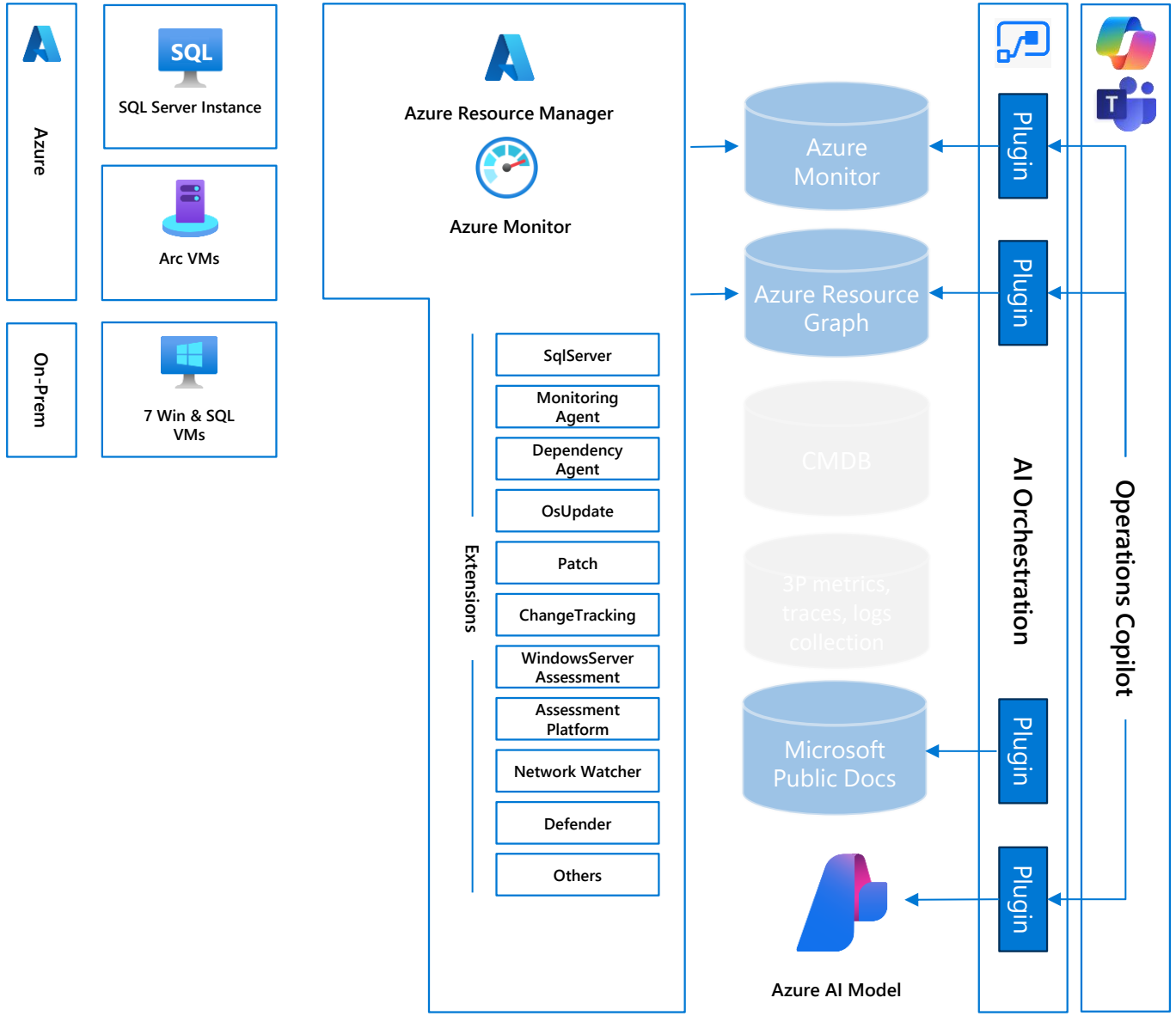
The best match found for the missing object is 'CN=Global Settings, CN=DC01, CN=Microsoft Exchange, CN=Services, CN=Configuration, DC=Contoso, DC=it.

Possible causes of the error could be:

1. The object was accidentally or mistakenly moved
2. There might be a replication issue between the domain controllers

...

Scenario demo AIOps



Real-world scenarios - Demo

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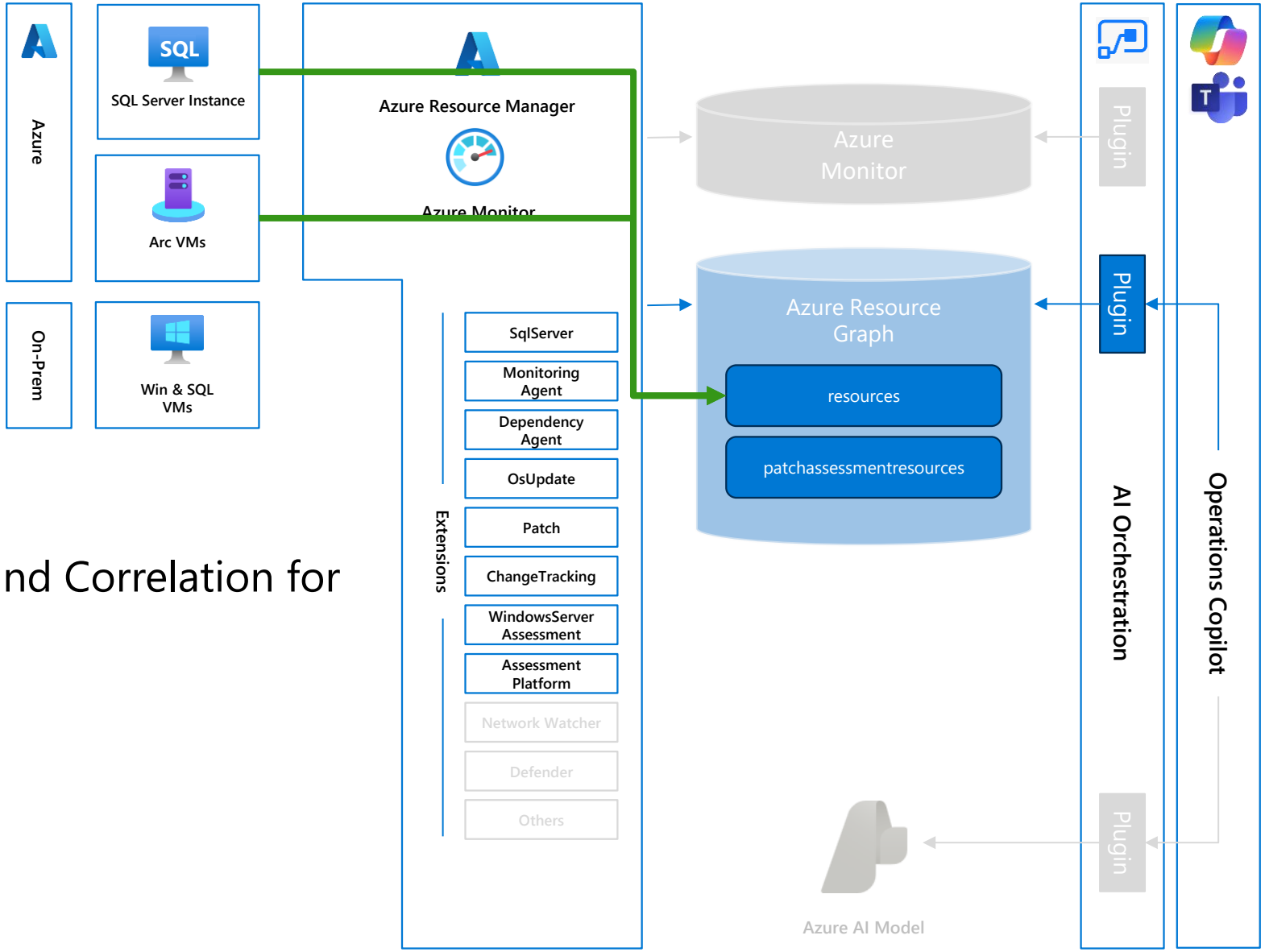
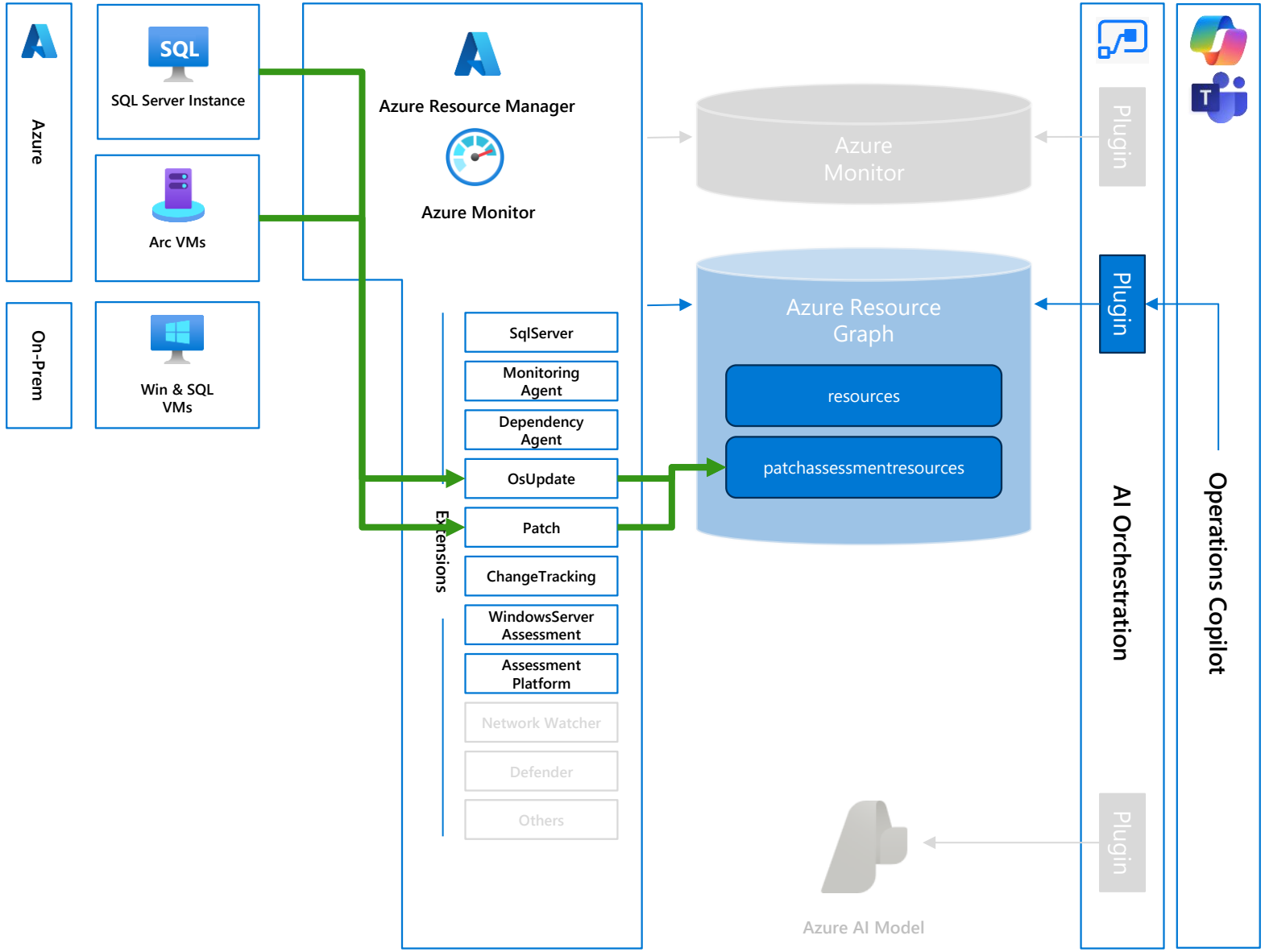


Use-Cases:

- Metadata Search and Correlation for Servers

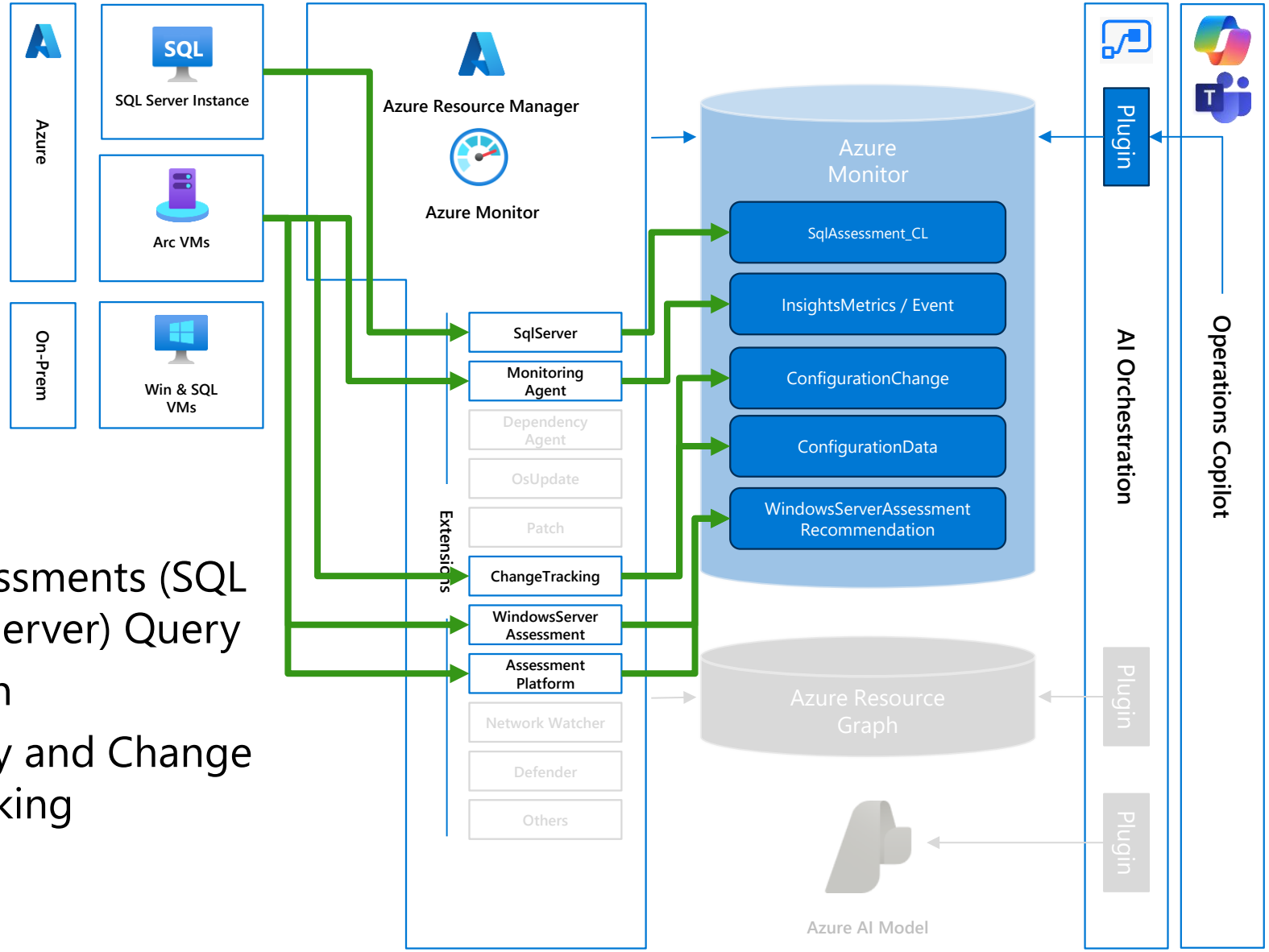
Scenario demo AIOps



<u>Use-Cases</u>:

- Patch Assessment

Scenario demo AIOps



Use-Cases:

- Best Practice Assessments (SQL Server, Windows Server) Query
- Anomaly detection
- Software Inventory and Change configuration tracking

Domande e Feedbaks

Microsoft

Principali processi di IT Operations secondo ITIL

Processo ITIL	Fasi del Processo (con descrizione)
Event Management (Rilevare e gestire eventi per prevenire problemi IT)	1. **Rilevamento evento**: raccolta automatica di eventi da sistemi, applicazioni e strumenti di monitoraggio. 2. **Registrazione**: log dell'evento in un sistema centralizzato. 3. **Filtraggio e correlazione**: eliminazione dei falsi positivi e raggruppamento di eventi correlati. 4. **Analisi**: valutazione della gravità e dell'impatto. 5. **Risposta**: azione automatica o manuale (es. invio alert, apertura ticket). 6. **Chiusura**: documentazione e archiviazione dell'evento.
Incident Management (Ripristinare il servizio il più rapidamente possibile)	1. **Identificazione**: rilevamento di un'interruzione o degrado del servizio. 2. **Registrazione**: creazione di un ticket. 3. **Categorizzazione**: classificazione per tipo di incidente. 4. **Prioritizzazione**: valutazione dell'urgenza e impatto. 5. **Diagnosi iniziale**: tentativo di risoluzione da parte del 1° livello. 6. **Escalation**: passaggio a livelli superiori se necessario. 7. **Risoluzione e ripristino**: soluzione del problema e ripristino del servizio. 8. **Chiusura**: conferma con l'utente e chiusura del ticket.
Request Fulfillment (Gestire richieste utenti in modo rapido ed efficiente)	1. **Ricezione richiesta**: da portale self-service o help desk. 2. **Validazione**: verifica che la richiesta sia legittima. 3. **Approvazione**: se necessaria, da parte di un responsabile. 4. **Esecuzione**: fornitura del servizio richiesto. 5. **Chiusura**: conferma con l'utente e archiviazione.
Problem Management (Identificare e rimuovere la causa radice di una serie di incidenti simili)	1. **Identificazione problema**: analisi di incidenti ricorrenti o gravi. 2. **Registrazione**: apertura di un ticket problema. 3. **Categorizzazione**: classificazione per tipo e impatto. 4. **Analisi causa radice (RCA)**: individuazione della causa principale. 5. **Creazione workaround o soluzione definitiva**: per ridurre o eliminare l'impatto. 6. **Chiusura**: documentazione e aggiornamento della knowledge base.
Access Management (Garantire accesso sicuro solo agli utenti autorizzati)	1. **Richiesta accesso**: da parte di un utente o sistema. 2. **Verifica autorizzazione**: controllo delle policy di sicurezza. 3. **Concessione o negazione accesso**: in base ai permessi. 4. **Monitoraggio utilizzo**: verifica dell'uso corretto. 5. **Revoca accesso**: quando non più necessario o in caso di violazioni.